KAIZEN MATHS

Lesson starter

Motion Graphs and Constant Acceleration

Speed and Acceleration

Name _____

SA1

A runner travels 150 m in 15 s. Find the average speed.

Show your working

KAIZEN MATHS

Lesson starter

Motion Graphs and Constant Acceleration

Speed and Acceleration

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SB1

A vehicle changes speed from 4 m/s to 12 m/s in 4 s. Find the acceleration and state whether the vehicle is accelerating or decelerating.

Show your working

KAIZEN MATHS

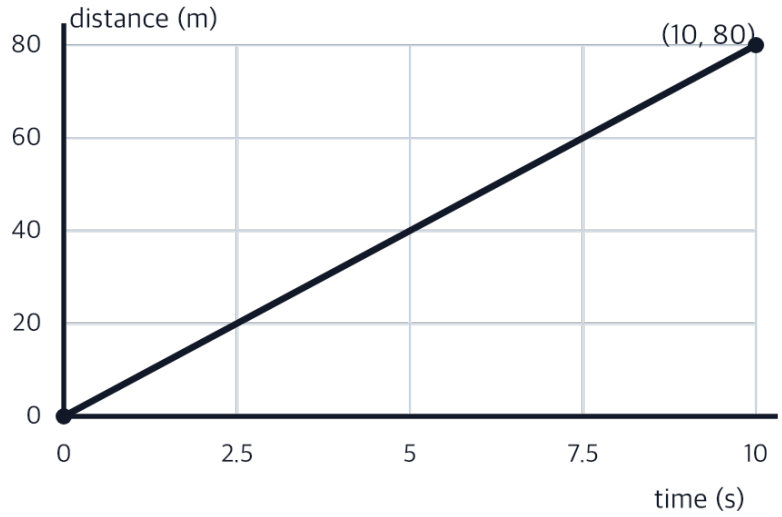
Lesson starter

Motion Graphs and Constant Acceleration

Speed and Acceleration

Name _____

SC1



Use the distance-time graph to find the speed.

MEC-50A8-L1 · SC

Show your working

KAIZEN MATHS

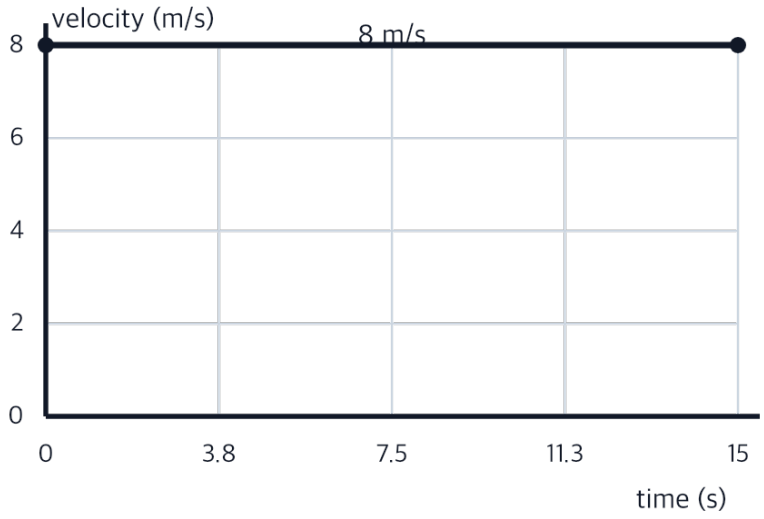
Lesson starter

Motion Graphs and Constant Acceleration

Speed and Acceleration

Name _____

SD1



Use the velocity-time graph to find the distance travelled.

MEC-50A8-L1 · SD

Show your working

KAIZEN MATHS

Exit ticket

Motion Graphs and Constant Acceleration

Speed and Acceleration

Name _____

EA1

A runner travels 280 m in 20 s. Find the average speed.

Show your working

KAIZEN MATHS

Exit ticket

Motion Graphs and Constant Acceleration

Speed and Acceleration

Name _____

EB1

A vehicle changes speed from 9 m/s to 33 m/s in 8 s. Find the acceleration and state whether the vehicle is accelerating or decelerating.

Show your working

KAIZEN MATHS

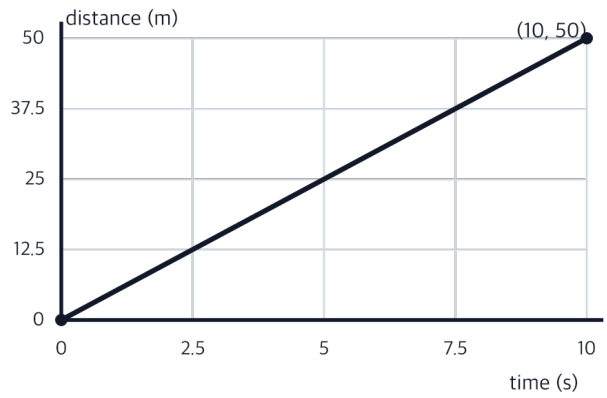
Exit ticket

Motion Graphs and Constant Acceleration

Speed and Acceleration

Name _____

EC1



Use the distance-time graph to find the speed.

Show your working

KAIZEN MATHS

Exit ticket

Motion Graphs and Constant Acceleration

Speed and Acceleration

Name _____

EA1

A runner travels 280 m in 20 s. Find the average speed.

Show your working

Teacher answers & working

Motion Graphs and Constant Acceleration · Speed and Acceleration

MEC-50A8-L1 · Student pages 1-3. Print at 100%, single-sided. Cut on dashed lines.

SA1

A runner travels 150 m in 15 s. Find the average speed.

Answer:

$10\ \text{m/s}$

Working:

Write the formula for average speed.

average speed = distance ÷ time

Substitute the distance and time.

average speed = 150 ÷ 15

average speed = 10

The average speed is 10 m/s.

SB1

A vehicle changes speed from 4 m/s to 12 m/s in 4 s. Find the acceleration and state whether the vehicle is accelerating or decelerating.

Answer:

$2\ \text{m/s}^2$

Working:

Write the acceleration formula.

$a = (v - u) \div t$

Substitute the initial speed, final speed, and time.

$a = (12 - 4) \div 4$

$a = 8 \div 4 = 2$

Interpret the sign of the acceleration.

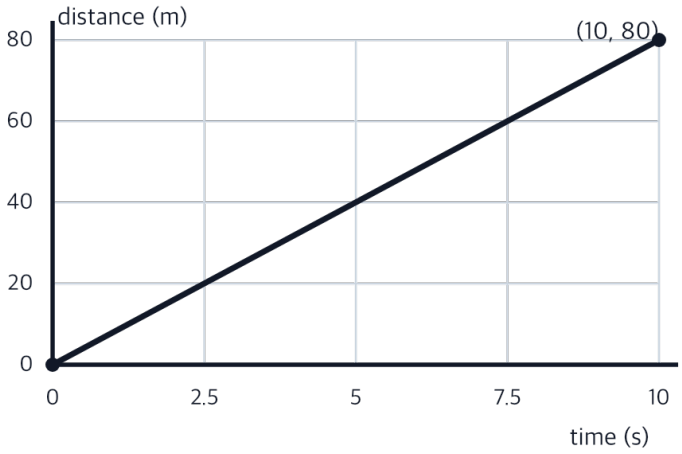
The acceleration is 2 m/s², so the vehicle is accelerating.

Teacher answers & working

Motion Graphs and Constant Acceleration · Speed and Acceleration

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SC1



Use the distance-time graph to find the speed.

Answer:

8 m/s

Working:

On a distance-time graph, speed is the gradient.

speed = change in distance ÷ change in time

Read two points from the graph.

from (0, 0) to (10, 80)

speed = 80 ÷ 10 = 8

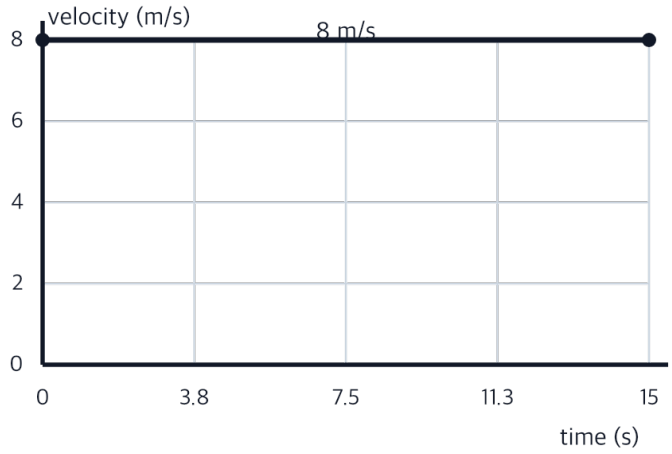
The speed is 8 m/s.

Teacher answers & working

Motion Graphs and Constant Acceleration · Speed and Acceleration

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SD1



Use the velocity-time graph to find the distance travelled.

Answer:

120 m

Working:

On a velocity-time graph, distance travelled is the area under the graph.

The area is a rectangle.

area = base × height

distance = 15 × 8 = 120

The distance travelled is 120 m.

EA1

A runner travels 280 m in 20 s. Find the average speed.

Answer:

14 m/s

Working:

Write the formula for average speed.

average speed = distance ÷ time

Substitute the distance and time.

average speed = 280 ÷ 20

average speed = 14

The average speed is 14 m/s.

Teacher answers & working

Motion Graphs and Constant Acceleration · Speed and Acceleration

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EB1

A vehicle changes speed from 9 m/s to 33 m/s in 8 s. Find the acceleration and state whether the vehicle is accelerating or decelerating.

Answer:

$3\ \text{m/s}^2$

Working:

Write the acceleration formula.

$a = (v - u) \div t$

Substitute the initial speed, final speed, and time.

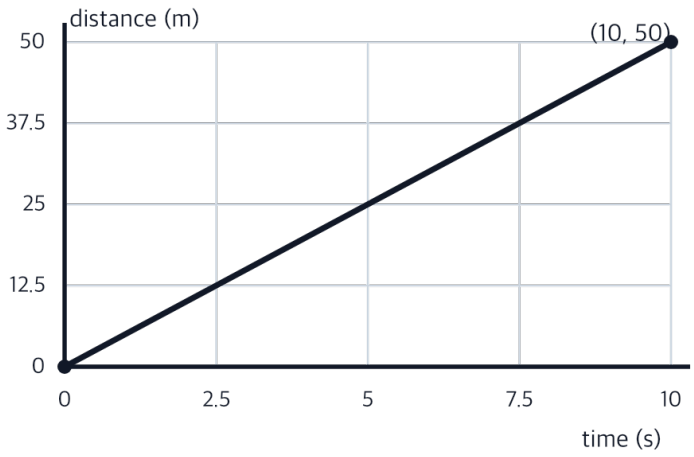
$a = (33 - 9) \div 8$

$a = 24 \div 8 = 3$

Interpret the sign of the acceleration.

The acceleration is 3 m/s², so the vehicle is accelerating.

EC1



Use the distance-time graph to find the speed.

Answer:

$5\ \text{m/s}$

Working:

On a distance-time graph, speed is the gradient.

speed = change in distance ÷ change in time

Read two points from the graph.

from (0, 0) to (10, 50)

speed = 50 ÷ 10 = 5

The speed is 5 m/s.